GROUP 9 · EC8208 · SOFTWARE ARCHITECTURE

REOAGMS

Renewable Energy Operations, Asset and Grid Management System

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One gateway, six independent services

React console · nginx · live updates over WebSocket

API Gateway · JWT validation · role policy

Identity	Asset	Monitoring	Maintenance	Notification	Analytics
PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL	PostgreSQL

RabbitMQ events · Redis cache · Prometheus and Grafana

LIVE Console: sign in, Control room

The registry every service relies on

Facility
5 types

›

Asset
7 types

›

Sensor
9 metrics

5 · 11 · 32

facilities · assets · sensors

Server-side

search and paging

Validated

every form

LIVE Assets page

From a sensor reading to an instant alert

Reading
every few seconds

›

Rule check
threshold

›

Alert
one per asset

›

RabbitMQ
event

16,000+
readings stored

7
alert rules

10 s
Redis cached dashboard

LIVE Live monitoring page

Faults become tracked work automatically

Alert event high or critical › **Work order** created › **Technician** assigned › **Completed** in history

0 clicks
to open a corrective order

4
technicians

1
active order per asset

LIVE Maintenance board

Every alert delivered and recorded

Queue
durable

›

Consumer
schema check

›

Channels
email · push

›

History
stored

Instant
in-app push

3
automatic retries

100%
deliveries recorded

LIVE Notifications page

Trustworthy energy figures and reports

Source APIs — REST only › **Calculation** — power to kWh › **Reports** — stored › **CSV** — export

kWh
integrated from power

6
KPIs with targets

5 + 5
energy and performance reports

LIVE Analytics page

Five roles, enforced at the gateway

Administrator	Everything, users and audit log
Operations Manager	Assets, rules, work orders, reports
Plant Operator	Monitor and handle alerts
Maintenance Engineer	Assign and complete work orders
Executive	Read-only dashboards and reports

LIVE executive sign-in, read-only

Observable, tested, one command to run

60
test methods

7
services monitored

19
containers

1
command to start

LIVE Grafana platform health

Thank you

Questions and discussion

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